

THE WORLD ENGINE

A New Computational Framework for Predictive Discovery and Latent Behavioral Dynamics

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Research paper | August 2026

Digital discovery systems have become highly effective at predicting actions from language, interaction history, and learned user representations. We herein propose a different generation of discovery system, which we call the World Engine. It models behavior as the observable endpoint of an evolving process and asks whether shared behavioral dynamics, current signals, and explicit state transitions can reveal what may become relevant before a user has expressed it through search or repeated consumption.

The framework is intended for social feeds, streaming services, programmatic advertising, digital commerce, and other environments in which a system selects the next piece of information presented to a person. A social application organized around visual statements and questions, bounded response options, voting, and short-lived conversations provides the first environment in which the hypothesis can be implemented and tested.

Abstract

Modern discovery systems learn powerful representations from language, content, interaction sequences, and population behavior. Their predictions remain anchored in observable digital traces, however, while the cognitive and contextual processes that generate those traces are only indirectly represented. This paper introduces the World Engine, a computational framework for a new class of predictive discovery systems. The framework treats behavior as a probabilistic emission from an evolving, partially observed state. It organizes shared behavioral dynamics as Archetypes, instantiates alternative transition models called Digital Minds, detects emerging multivariate patterns through Signal Intelligence, expresses those models through synthetic identities, maps real users to mixtures of Digital Minds, and evaluates candidate information by simulating possible state transitions. The approach draws on representation learning, sequential recommendation, state-space models, POMDPs, world models, JEPA, behavioral science, memory, attention, and decision neuroscience. These connections motivate the architecture without implying that a Digital Mind is a biological replica or that private mental states can be read from platform data. We report a working synthetic-world prototype that implements the representational hierarchy and content-to-identity execution path, while distinguishing those engineering results from unvalidated claims about human latent state. The central empirical claim is narrower: explicit transition structure may reduce dependence on long individual histories and improve prediction under cold start, context change, preference shift, and novel discovery. We specify prospective tests and identify conditions under which the framework should be rejected or reduced to simpler models.

Keywords: predictive discovery; latent behavioral state; signal intelligence; digital minds; world models; recommender systems; POMDP; behavioral dynamics

1. Introduction: from history to predictive discovery

The dominant architecture of digital discovery begins with traces of prior behavior. A system observes searches, views, watch time, skips, purchases, follows, comments, and other actions. It represents the available content through text, images, audio, metadata, entities, and learned embeddings. It then estimates which candidate is most likely to produce a desired response from a user or account. The methods have changed dramatically, from hand-built rules and collaborative filtering to multimodal foundation models, graph learning, sequential transformers, and reinforcement learning. The organizing direction remains broadly similar: infer the next action from what has already become digitally observable.

This paper advances a different thesis. The next generation of discovery systems should not be defined only by a better score over the same observation model. They should represent the evolving process from which an action emerges. In the physical world, behavior is shaped by interacting constraints and processes: physiology, attention, memory, active goals, affect, social context, learned habits, environmental cues, and action preparation. A digital service ordinarily observes only the terminal trace, such as a click, vote, pause, or purchase. The World Engine asks whether a structured model of the hidden process can support a form of discovery that is less dependent on accumulated history and more capable of predicting relevance before it has been explicitly expressed.

The proposal is aligned with a broader movement in artificial intelligence from surface prediction toward world models. LeCun argues that an intelligent agent requires a model able to predict possible future world states as a function of imagined actions (LeCun, 2022). Model-based reinforcement learning similarly separates state estimation, transition prediction, and planning. The World Engine applies this orientation to information discovery. The world to be modeled is not physical mechanics alone. It includes the behavioral field created by people, content, context, and the platform policy that determines what each person is allowed to observe.

The biological analogy must be used with discipline. The framework does not claim to measure neurotransmitters, reconstruct a brain, determine free will, or reveal a true private intention. Neuroscience and psychology provide evidence that overt action is preceded and shaped by processes that unfold over time. They also show that attention, memory, valuation, motivation, and context interact rather than act as isolated labels. These observations support a computational design principle: a prediction system may benefit from representing multiple latent processes and their transitions. The validity of that design must be established by prediction, calibration, intervention, and falsification, not by the authority of a biological metaphor.

The intended scope extends beyond social media. Any service that decides which information, offer, advertisement, product, song, video, question, or recommendation appears next faces the same general problem. A streaming service may repeat yesterday's preference when a listener wants a different mood. A programmatic advertising system may continue displaying flights after the flight has already been purchased. A commerce platform may infer a stable product interest from one accidental session. These failures are not proof that current systems are primitive. They reveal a structural challenge: the same digital trace can be generated by different states, and a change in state may occur before a new trace announces it.

The World Engine is presented as a theoretical model, a working synthetic-world prototype, and a research program. Its first controlled application is a social environment organized around visual statements and questions, bounded answer options, voting, and short-lived sequences of interaction. This setting makes behavioral signals unusually explicit and creates opportunities for randomized testing. The paper defines the architecture, explains how Archetypes generate Digital Minds and identities, describes what has and has not been implemented, specifies how Digital Minds may be mapped to real users, and proposes experiments that can determine whether the approach offers more than a renamed sequence model.

How current discovery systems construct relevance

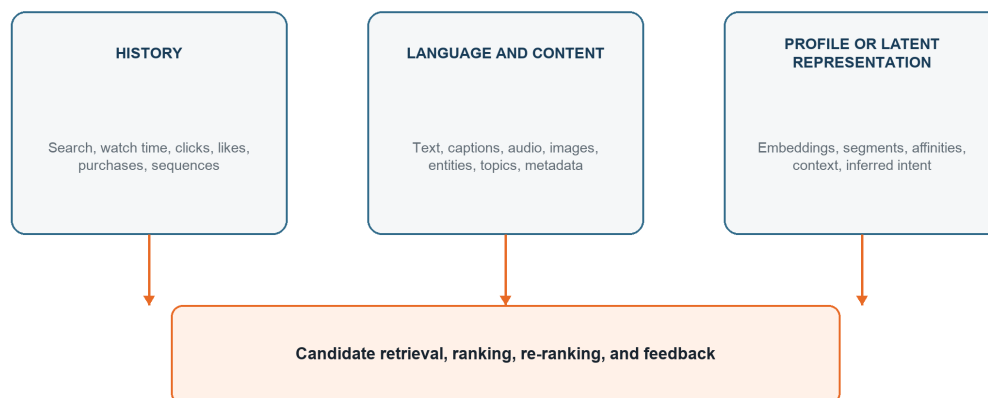


Figure 1. Contemporary discovery systems combine much more than keywords. History, multimodal content, context, and learned user representations enter a multistage retrieval and ranking process. The World Engine changes the factorization of this evidence rather than denying its value.

2. Scientific motivation and the observability problem

The distinction between observation and state is classical in statistics and control. In a hidden Markov model or partially observable Markov decision process, the state cannot be observed directly. The system maintains a belief distribution and updates it as new evidence arrives (Kaelbling, Littman, and

Cassandra, 1998). Predictive state representations define state through predictions of future observable tests instead of assuming that one privileged hidden ontology can be recovered (Littman, Sutton, and Singh, 2002). These traditions provide the formal foundation for the World Engine. A Digital Mind is best understood as a predictive state model with named computational interfaces, not as a claim about the complete inner condition of a person.

The human case introduces a pronounced measurement asymmetry. In the physical organism, action emerges from processes distributed across the nervous system and body. Sensory input changes neural activity; memory and learned associations alter interpretation; homeostatic and affective variables change value; attention gates what receives processing; goals and context constrain action. Some of these processes can be studied with neuroimaging, electrophysiology, psychophysics, physiological measurement, and controlled experiments. They are not continuously observed by the discovery systems used in ordinary digital services.

Digital systems therefore receive a radically compressed projection of the process. They see that a user paused for four seconds, selected the first option, ignored an advertisement, replayed a song, or left the application. They usually do not observe whether the action reflected attraction, fatigue, confusion, habit, social pressure, task completion, accidental exposure, or a transient change in goal. Even an expressive embedding can entangle these explanations because the training objective rewards accurate outcomes rather than a unique decomposition of the process.

This is the physical-to-digital observability gap. It does not imply that biological measurement must be added to a consumer platform. Such measurement would often be impractical, intrusive, and ethically unacceptable. The engineering question is whether platform events contain enough indirect structure to maintain a useful posterior over behaviorally relevant state. If the answer is yes, the system can model uncertainty over several plausible processes. If the answer is no, the system should abstain and revert to a simpler, more auditable policy.

Behavioral science suggests why state can matter. Preferences change with framing and reference points (Kahneman and Tversky, 1979; Tversky and Kahneman, 1981). Visceral and motivational states alter valuation (Loewenstein, 1996). Working, episodic, and semantic memory contribute differently to present action (Tulving, 1972; Baddeley, 2000). Goal-directed and stimulus-driven attention rely on partly distinct control systems (Corbetta and Shulman, 2002). Decision neuroscience describes value construction and reward-prediction errors as distributed processes rather than fixed preference lookups (Schultz, Dayan, and Montague, 1997; Rangel, Camerer, and Montague, 2008). The computational implication is that stable identity, active goal, retrieved memory, salience, and context should not be forced into one update rate.

Research on voluntary action requires particular restraint. Readiness potentials and above-chance decoding before reported awareness show that measurable processes can precede overt action (Libet et al., 1983; Soon et al., 2008). Their interpretation remains contested. Stochastic accumulator models can produce a readiness-potential-like average without a settled unconscious decision, and later reviews caution against treating early activity as proof that action was already fixed (Schurger, Sitt, and Dehaene, 2012; Schurger et al., 2021). The defensible premise is not that neural activity determines a choice before consciousness. It is that overt action is the endpoint of processes that evolve over time, remain uncertain, and can sometimes be measured before execution.

Predictive processing and active inference offer influential theories of hierarchical prediction and uncertainty (Rao and Ballard, 1999; Friston, 2010). The World Engine does not adopt the free-energy principle as its optimization objective. It borrows three operational ideas: state estimates should carry uncertainty, prediction errors should revise them, and actions change the observations available in the future.

The observability gap between physical process and digital trace

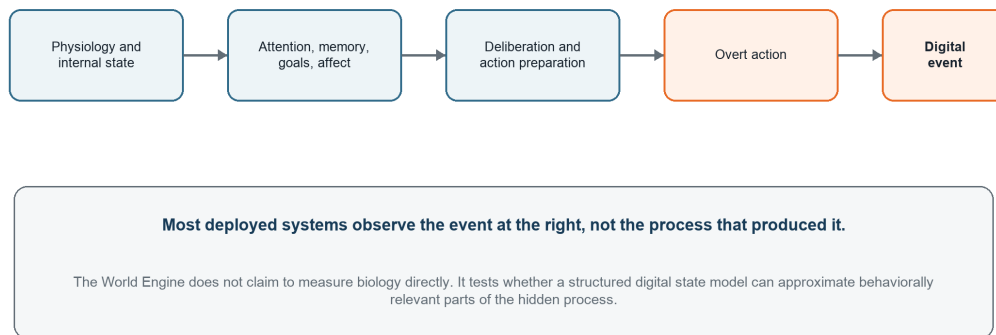


Figure 2. The physical-to-digital observability gap. Ordinary discovery systems observe a compressed event after a longer process has unfolded. The proposed model approximates only those hidden dynamics that improve prospective behavioral prediction.

3. The World Engine hypothesis

Let $o_{1:t}$ denote the observations associated with an identity up to time t , c_t the measured context, at a candidate intervention, and y_{t+1} the next observable response. A strong sequential recommender may learn $P(y_{t+1} | o_{1:t}, c_t)$ directly. The World Engine introduces a latent state z_t , a belief $q(z_t | o_{1:t}, c_{1:t})$, a Signal representation $s(a_t)$, and an action-conditioned transition model $p(z_{t+1} | z_t, s(a_t), W_t)$. The predicted response is emitted from the successor state rather than scored directly from the historical sequence.

The first part of the hypothesis concerns representation. A recursively updated belief over typed, multi-timescale state may preserve decision-relevant information more effectively than a single undifferentiated history vector when context, goal, and preference change at different rates. Stable dispositions should not be overwritten by one unusual session. A temporary goal should not be diluted across years of history. A context-specific action should not automatically become an enduring identity label.

The second part concerns transfer. Shared behavioral dynamics can be learned across identities and represented through Archetypes and Digital Minds. A new or sparsely observed user can then be mapped to a distribution over existing transition models. The claim is not literal independence from data. Population data created the models, and current observations update the mapping. The testable claim is reduced dependence on a long identity-specific record.

The third part concerns discovery. A candidate should be evaluated by the state transition it may induce, including uncertainty and delayed consequences. The system can therefore consider information that is semantically distant from a user's recorded history when the present state and shared transition model make it relevant. A football question may be appropriate for a traveler who has never followed football online; a cultural question may be relevant to an account dominated by animal videos; a new music genre may be welcomed in one context and rejected in another. These are predictions to be tested, not evidence that an unexpressed preference has been read from the brain.

The fourth part concerns the platform as part of the modeled world. Recommendation changes exposure, exposure changes behavior, and behavior changes the data used to infer the user. A system can create the pattern it later interprets as preference. World State W_t therefore includes identities,

content, relations, shared context, and the policy that controls observation. This permits evaluation of feedback loops, creator exposure, coordination, and concentration.

The hypothesis is falsifiable. Typed state must outperform capacity-matched sequence models under the conditions for which it is proposed. Archetypes must improve cold-start adaptation without hardening demographic stereotypes. Signal Intelligence must detect prospective structure beyond popularity, semantics, and exposure artifacts. Candidate rollouts must improve calibrated long-horizon outcomes. If these conditions are not met, the additional structure should be removed.

$$q(z_t | o_{1:t}, c_{1:t}) \rightarrow p(z_{t+1} | z_t, s(at), W_t) \rightarrow p(y_{t+1} | z_{t+1})$$

Implemented prototype: a synthetic behavioral world

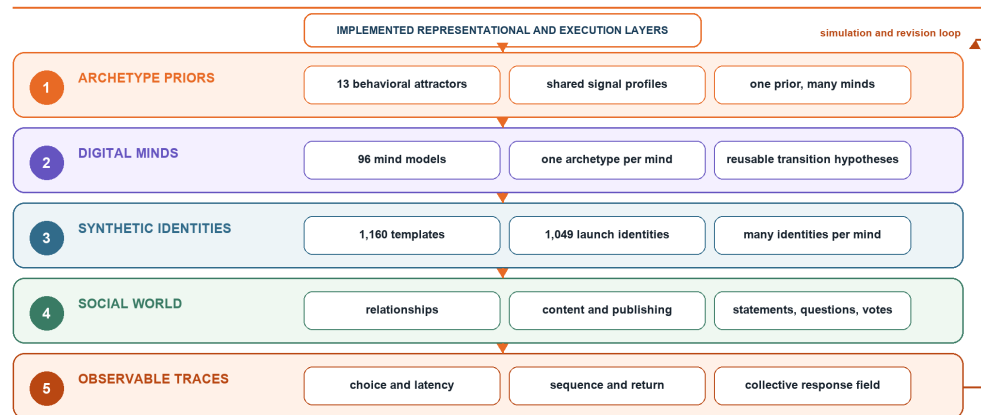


Figure 3. Implemented prototype hierarchy. Thirteen Archetype priors organize 96 Digital Minds, which are expressed through a synthetic population and operated inside a content, publishing, relationship, and interaction environment. Counts describe the August 2026 prototype and are not empirical evidence of predictive validity.

4. Why present discovery systems reach a representational limit

A credible argument for a new paradigm must address the strongest existing systems. Industrial recommendation is usually multistage. Candidate generation retrieves a manageable set from a large corpus. Ranking predicts several possible outcomes. Re-ranking applies diversity, freshness, policy, and business constraints. Exploration allocates some exposure to uncertain alternatives. Candidate generators and rankers use collaborative structure, multimodal encoders, graph neighborhoods, sequences, context, and learned embeddings (Covington, Adams, and Sargin, 2016; Cheng et al., 2016; Wang et al., 2019). These systems are not simple keyword engines.

Sequential models already address short-term variation. GRU4Rec, SASRec, transformers, and selective state-space models encode ordered interactions and allow recent events to change the effective representation (Hidasi et al., 2016; Kang and McAuley, 2018; Gu and Dao, 2023). Latent-intent recommenders place a probabilistic variable between sequence and policy and have reported gains in large-scale experiments (Chang et al., 2022). Reinforcement-learning recommenders optimize slates or trajectories rather than one isolated click (Shani, Heckerman, and Brafman, 2005; le et al., 2019). Satisfaction research supplements engagement with surveys and long-term behavioral surrogates (Christakopoulou et al., 2021; Xu et al., 2022).

The proposed limit is therefore specific. These systems are optimized primarily through behavioral emissions. Their internal representations can be extremely rich while remaining entangled. They need not expose whether one dimension reflects an active goal, context, position bias, repeated exposure, memory retrieval, or a stable preference. For ordinary in-distribution ranking, that distinction may be unnecessary. It becomes important when the policy changes, history becomes misleading, explanation matters, or the system must predict response to a candidate with little semantic or collaborative connection to prior behavior.

The cat-lover example illustrates the issue in simple terms. An account watches or likes fifteen pieces of content containing cats. The sixteenth cat item is a reasonable candidate because recent behavior and collaborative neighbors support it. The system may also infer affinities with pet care, wildlife, or the musical Cats through semantic and population structure. What it cannot observe directly is whether another cat item is rewarding now. The user may be satiated, distracted, seeking novelty, or opening the application for a different purpose. Repetition can increase short-term certainty while reducing attention or satisfaction.

The same problem appears in programmatic advertising. A person searches for a flight, purchases it, and continues to receive advertisements for the same route. The observable signal indicated commercial intent, but the purchase changed the state from searching to completed. A funnel model may shift toward hotels or rental cars, yet those categories can also be wrong if the traveler is staying with friends and using public transport. The next relevant item may be a football match, an exhibition, or information that has no recorded association with the transaction. Continued exposure to obsolete offers can produce avoidance, irritation, and lower trust. The system has optimized a proxy after the underlying task changed.

It is tempting to describe the failure in biological language and say that irrelevant content produces negative hormones while relevant content rewards the brain. The underlying direction is reasonable, but the scientific formulation must be more careful. Content can alter attention, affect, arousal, valuation, and approach or avoidance behavior. Consumer platforms rarely measure those processes directly. They can measure proxies such as dwell, response latency, abandonment, repeated return, voluntary satisfaction, and controlled preference reports. The World Engine seeks a model whose predictions align better with those outcomes without pretending that a click reveals a neurotransmitter.

The need for a new paradigm follows from this mismatch. Discovery is not only the retrieval of items similar to a recorded profile. It is the selection of information capable of becoming relevant to a changing person in a changing context. A system that models transition can ask a different question: given the current uncertain state, what candidate is likely to produce interest, understanding, satisfaction, or another explicitly chosen outcome?

5. Computational framework

A World State W_t contains a population of Digital Minds, identities, content and other external entities, a time-varying relation graph, shared context, and the deployed policy. For identity i , the Digital Mind state separates slowly varying constraints from dynamic state, memory, active goals, and measured context. The operational object is a posterior distribution over this tuple, not a deterministic label.

Functions are reusable computational operators through which Signals change state. Candidate Functions include attention allocation, memory retrieval, novelty response, value estimation, goal activation, social influence, inhibition, and context gating. Their names describe roles in the model. They do not claim one-to-one correspondence with neural regions. A Function may be implemented with a

gated transition, transformer block, selective state-space layer, neural differential equation, mixture of experts, or a structured probabilistic kernel.

Memory is not an undifferentiated log. Short-lived traces can decay quickly. Episodic summaries preserve event identity and context. Semantic or habitual components change slowly. Retrieval is conditioned on current state and candidate Signal, allowing the same stored event to influence different decisions in different ways. Goals are temporary control variables with uncertainty and explicit expiration. Context contains measured external conditions such as time, device class, task, coarse location, social setting, and availability, subject to lawful and proportionate collection.

World State makes relational influence explicit. One identity's action can change the information, perceived norm, or opportunity available to another. Content creators respond to exposure, users coordinate or diverge, and organizations change incentives. The environment is therefore nonstationary even when individual parameters remain fixed.

$$W_t = (\{M_{i,t}\}, E_t, G_t, C_t, \text{policy}_t)$$

$$z_{i,t} = (b_i, x_{i,t}, m_{i,t}, g_{i,t}, c_{i,t}), \quad q(z_{i,t}) = \text{belief over slow constraints, dynamic state, memory, goals, and context}$$

$$q(z_{i,t} \mid D_{i,1:t}) \propto E(o_{i,t} \mid z_{i,t}) \int T(z_{i,t} \mid z_{i,t-1}, s_{i,t}, W_t) q(z_{i,t-1}) dz_{i,t-1}$$

Signals interact with Functions to update a Digital Mind

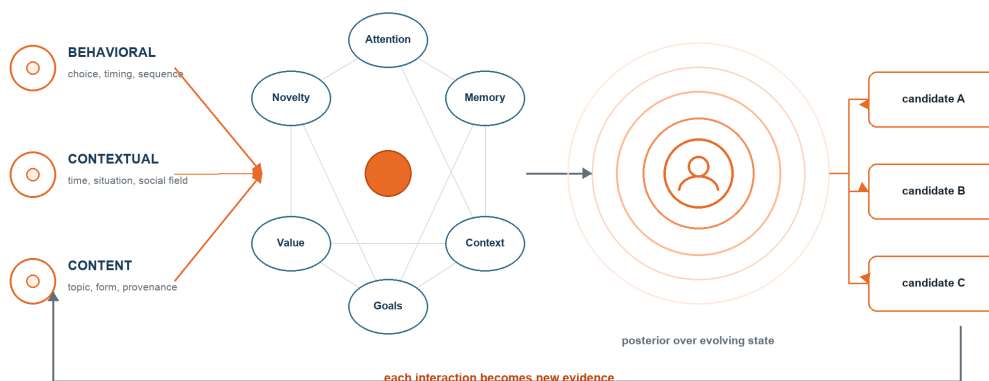


Figure 4. Signals enter an interacting functional network and update a posterior over evolving state. Candidate outcomes are conditioned on that state, and each observed interaction returns as new evidence.

5.1 Signal Intelligence

Signal Intelligence is not another name for a trending list. A conventional trend often identifies an entity whose volume or growth exceeds a baseline: a phrase, account, song, location, or item becomes visible because many interactions reference it. A Signal is any reproducible pattern in the joint event process that carries predictive information about a state transition or future action within a specified population and time window.

The object may be semantic, but it may also be behavioral. People may increasingly choose the first answer regardless of topic. A subgroup may consistently oppose the local majority. Response time may shorten after one sequence and lengthen after another. A geographic cluster may change before the wider population. A series of questions may work because of its order even when the questions share

no keywords. A user may deviate sharply from the behavior predicted by the currently dominant Digital Mind. Each of these patterns can become a Signal.

Let $e_t = (i, a, v, \tau, l, p, x_i)$ denote an event produced by identity i on candidate a , with response v , time τ , coarse location l , exposure propensity p , and lawful context x_i . A candidate pattern r is evaluated against a context-appropriate baseline B . Its strength depends on effect size, persistence, population specificity, novelty, predictive lift, and posterior confidence. Frequency alone is insufficient. A rare pattern may be highly informative for one Digital Mind while a viral item may convey little new information for another.

Signal provenance is essential. The system must distinguish content properties from answer position, exposure policy, social propagation, and interaction mechanics. Propensity correction or randomized exploration is required when the platform's own policy affects observation. False-discovery control is required because a sufficiently large stream contains accidental regularities. A Signal is promoted only when it predicts held-out future behavior or survives prospective testing. It is retired when its lift decays.

Signals operate at different scales. A burst may last minutes. A daily rhythm may recur for months. A stable interface bias may persist until the interface changes. Continuous-time models, change-point detection, temporal graphs, sequence mining, contrastive learning, and conditional mutual information offer implementation tools. The framework remains agnostic about the detector, but it requires an uncertainty estimate and an explicit half-life.

Routing connects Signal Intelligence to Digital Minds. The system estimates whether a Signal is expected to reduce uncertainty, activate a Function, or cause a useful transition under mind hypothesis k . This creates a sparse interaction structure between Signals, Functions, and Digital Minds. It does not produce one universal feed of what is globally popular.

$$S_{r,t} = (\text{intensity, uncertainty, persistence, novelty, lift, half-life})$$

$$I_{r,t} = I(R_{r,t}; Y_{i,t+1} | C_{i,t}, P_{i,t}) - \lambda_1 \text{FDR}_{r,t} - \lambda_2 \text{DriftRisk}_{r,t}$$

Signal Intelligence detects meaningful structure in the event field

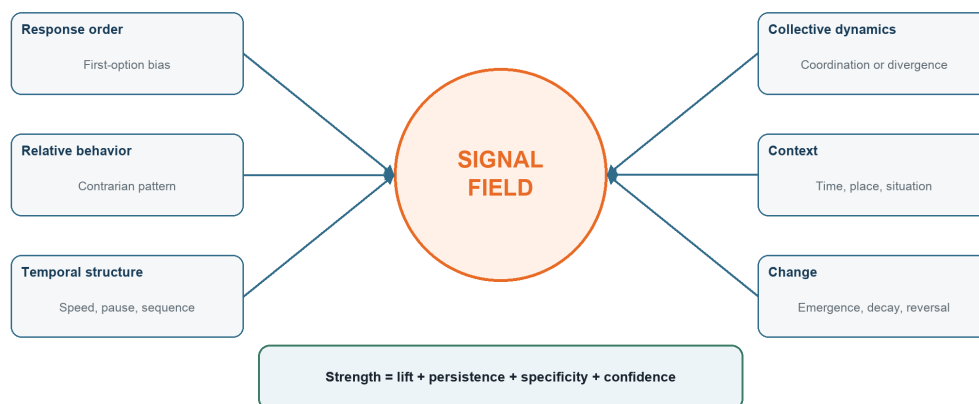


Figure 5. Signal Intelligence treats trends as patterns in a behavioral field. Entity popularity is one possible pattern among many.

5.2 Archetypes

The term Archetype is used in a restricted computational sense. Jungian archetypes, marketing personas, demographic segments, and broad personality types do not provide an adequate state model for discovery. Trait psychology offers stronger measurement traditions, but constructs such as

extraversion or conscientiousness are not automatically the variables required to predict interaction with a digital platform (McCrae and Costa, 1997). In the World Engine, an Archetype is a hierarchical prior over transition dynamics.

An Archetype may encode shared tendencies in novelty response, speed of habituation, deliberation before voting, sensitivity to social proof, preference for coherent sequences, tolerance for informational discontinuity, or persistence of an active goal. It does not encode a topic preference such as football, cats, politics, or fashion. Two identities can share exploratory dynamics while caring about different subjects. Two identities can consume the same subject through different dynamics.

In the proposed learned system, Archetypes are estimated from behavioral sequences and intervention responses. In the current prototype they remain designed priors whose assumptions must be tested. They are not assigned from race, ethnicity, religion, gender, sexual orientation, nationality, or another protected characteristic. Behavior-only clustering can still reconstruct sensitive attributes, so the model must be probed for leakage and disparity. The posterior must be able to override the prior, and the system must permit expiry, correction, and abstention. A prior that cannot be revised is a stereotype, not a valid Archetype.

The number of Archetypes is selected through predictive criteria. Too few components collapse meaningful transition regimes. Too many components memorize individuals and recreate personas under another name. Held-out marginal likelihood, predictive clustering, nonparametric mixtures, and minimum-description-length criteria can guide complexity. Stability across time and random initialization must be reported.

$$p(\theta_i | D_i) \propto p(D_i | \theta_i) \sum_k p_k p(\theta_i | \text{Archetype } k)$$

$$\theta_i | h_k = \mu_{hk} + L_{hk} \epsilon_i, \quad \epsilon_i \text{ distributed as } N(0, I)$$

5.3 Digital Minds

An Archetype generates many Digital Minds by varying transition parameters and Function couplings. A Digital Mind represents one coherent hypothesis about how platform-relevant state changes under Signals and context. It contains a transition kernel, emission model, retrieval policy, and uncertainty model. It is not necessarily attached to one human account.

During simulation, many synthetic identities can be sampled from one Digital Mind to estimate the range of behavior implied by its parameters. During deployment, one real identity can maintain posterior mass over several Digital Minds. Conversely, one Digital Mind may explain observations from many identities. The relationship is many-to-many and changes as evidence arrives.

The same action can arise from different processes. A quick vote for the first option may reflect a position heuristic, low interest, time pressure, playful behavior, social imitation, or a genuine preference. The posterior should remain multimodal until another Signal discriminates among these explanations. A later contrarian vote, a change in latency, or a controlled answer-order manipulation can reweight the hypotheses.

Different histories can also converge to a similar present state. A long-time football viewer and a traveler with no recorded sports interest may both become receptive to information about a Barcelona match. The model does not infer that both identities have become football fans. It predicts that a candidate has state-dependent relevance at this moment.

$$M_k = (Z_k, T_k, E_k, F_k, R_k, U_k), \quad k \in \{1, \dots, K, \text{unknown}\}$$

5.4 Identities and situated expression

A Digital Mind is a reusable computational model; an identity is its situated expression inside a particular social world. The distinction is necessary because the same transition model can be expressed through different names, locations, relationships, content histories, styles, and social positions. These surface variables determine what the identity can observe and how its actions become visible, but they do not create a new Digital Mind whenever the presentation changes.

The implemented hierarchy is asymmetric. Each Digital Mind is linked to one Archetype prior, each Archetype can generate many Digital Minds, and one Digital Mind can support many synthetic identities. A real user may later carry posterior support across several Digital Minds. This prevents the identity record from becoming a persona that silently combines demographics, topics, and behavioral dynamics into one permanent label.

Synthetic identities make the model executable. They receive content, form relationships, publish statements or questions, vote, pause, skip, and produce sequences of events. Their purpose is not to imitate a named individual. They expose the consequences of the encoded assumptions and create a controlled environment in which transition models, routing rules, and collective dynamics can be inspected before human prediction claims are made.

6. Prototype implementation and synthetic world

A working prototype implements the principal representational and execution layers of the World Engine. At the August 2026 snapshot, the registry contained 13 Archetype priors and 96 active Digital Minds, with every Digital Mind linked to one Archetype and represented by at least one downstream identity. The identity system contained 1,160 templates and 1,049 mind-linked launch identities. These numbers describe software objects and population coverage. They are not experimental sample sizes, learned psychological categories, or evidence that the hierarchy predicts human behavior.

The prototype operates as a synthetic social world. Its Mind Engine maintains the Archetype-to-Digital-Mind registry. Its Identity Engine materializes those models as situated actors. Its Content Engine holds acquired or generated media. Its Publishing Engine assigns content to an identity and schedules an observable expression. Its World Engine maintains relationships, follow links, shared events, and the evolving environment in which identities encounter one another. This modular separation permits the same Digital Mind to be tested through multiple identity expressions and social contexts.

The present content path begins with acquisition into a media library. A source item, such as a short video of a skateboarder falling, is represented through available semantic metadata and current category or keyword rules. The system assigns the item to a compatible Digital Mind, selects a publishing identity associated with that mind, and can adapt the wording or presentation before publication. Content may also be generated. This path demonstrates content acquisition, routing, identity-based expression, and publication; it does not yet demonstrate autonomous inference of latent human state.

The application layer presents visual statements and questions with bounded response options. People can vote, skip, pause, create a moment, inspect community responses, and continue into a short-lived conversation. A single exposure can therefore yield a structured observation rather than a generic engagement count. The record can include the candidate and its provenance, answer order, selected option, latency, skip or abandonment, social evidence visible at response time, continuation, return, and coarse context. These observations are the raw material from which behavioral signals may later be validated.

Several parts of the current system remain deliberately simulated or rule-based. Archetypes, Digital Minds, their parameters, content routing, and much of identity behavior contain designed assumptions. Keyword and category features still contribute to assignment. The synthetic population can evolve through scheduled events, relationships, content flows, and accumulated traces, but this evolution should not be confused with an autonomously learned cognitive model. The prototype establishes that the proposed ontology can be implemented as an operating environment. It does not establish that the ontology is the uniquely correct factorization of human behavior.

This distinction defines the scientific boundary of the present work. Implemented means that the software object, relationship, or execution path exists. Simulated means that its behavior follows explicit assumptions that can be exercised and inspected. Proposed means that a learning or inference mechanism remains to be trained and prospectively validated. The transition from simulation to evidence requires mapping real users under uncertainty and showing predictive gains against strong baselines.

Object	Computational meaning	Prototype status
Archetype	Shared prior over transition dynamics	13 registered priors
Digital Mind	Reusable model linked to one Archetype	96 active and represented
Identity template	Specification for a situated synthetic actor	1,160 templates
Launch identity	Mind-linked identity instantiated for the synthetic world	1,049 identities
Real user	Human account whose mapping remains an empirical task	Not an identity template or validation cohort

Table 1. Prototype data dictionary. Counts refer to the inspected August 2026 implementation snapshot and should be versioned as the system changes.

6.1 Initial simulation observations

The synthetic world can be used as a controlled policy comparison. The same population, candidate reservoir, relationship graph, and event schedule are exposed to three delivery policies. A random policy samples eligible content without behavioral matching. A history, language, and persona policy favors semantic continuity and previously expressed affinities. The optimized World Engine policy routes modeled Signals toward compatible Digital Minds and selects candidates according to their predicted state transition. The current internal observation is qualitative; a formal simulation study should repeat the comparison across seeds and report its sensitivity to the encoded assumptions.

Within the designed simulation, the policies produce different qualitative trajectories. Random delivery progressively weakens voting intensity and browsing persistence because candidate sequences frequently fail to match the active synthetic state. The history and language policy produces an early voting increase and a slower browsing plateau, but loses interaction when repetition produces modeled saturation or when the active state changes. The optimized World Engine shows a rapid voting response, a delayed increase in browsing persistence after exploratory friction, and recovery of World State interaction potential after a simulated context shift. These differences arise because content is redirected when the Digital Mind model indicates saturation, context change, or a stronger competing Signal.

Voting intensity is defined as votes per eligible exposure rather than raw clicks. Browsing persistence summarizes continuation through successive content items or conversations. A population-level World State interaction-potential index aggregates the proportion of identities in interaction-ready states, candidate-to-state compatibility, and the propagation opportunity created by the relationship graph. This

quantity is not a measure of well-being or biology. It is an internal simulation variable used to examine whether the designed system remains behaviorally active under different policies.

These results are best understood as an executable sanity check. They show that the implemented assumptions generate the intended directional behavior: stronger Signals routed to compatible Digital Minds increase model-defined interaction, while random or repeatedly semantic delivery reduces it. Because the same assumptions help determine both the synthetic agents and the policy, the comparison cannot establish superiority for human users. It motivates, but does not replace, controlled user testing.

$$\text{Voting intensity } V_t = \text{votes at } t / \text{eligible exposures at } t$$

World State interaction potential $\text{OMEGAT} = \text{aggregate of state readiness, candidate compatibility, and relational propagation}$

Illustrative simulation output under three delivery policies

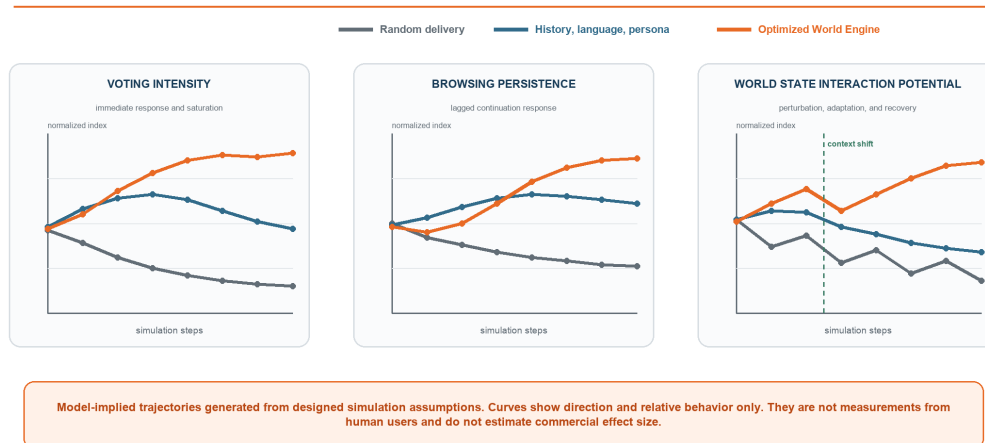


Figure 6. Illustrative synthetic policy comparison. Voting shows immediate response and saturation, browsing persistence changes more slowly, and World State interaction potential is perturbed by a context shift before adaptation. The optimized World Engine remains the strongest model-implied policy, but the three panels intentionally represent different dynamics. Curves are normalized trajectories, not human-user measurements or claimed effect sizes.

7. Mapping Digital Minds to real users

The bridge from theory to implementation is the identity-to-mind mapping. Without this bridge, Digital Minds remain a catalogue of plausible simulations. The mapping must answer two questions at every decision point. Which mind hypotheses currently explain the user's observations? What state does each hypothesis assign to the user now? Both answers carry uncertainty.

Let M_k define one Digital Mind and $\alpha_{i,t}(k)$ the posterior support that identity i is currently described by that hypothesis. Within each Digital Mind, $q_k(z_i, t | D_i, 1:t)$ represents the posterior over dynamic state. The predictive response to candidate a marginalizes across both sources of uncertainty. New evidence updates the state within each model and the support between models. The system never needs to declare that a user is one permanent mind.

Initial mapping can begin with a population prior. A small number of low-risk interactions, such as controlled questions with randomized order, provide information about response style, novelty tolerance, sequence sensitivity, and context. The system chooses some questions for relevance and others for information gain, subject to experience and safety constraints. This is related to preference elicitation

and contextual bandits, but the target is a transition model rather than a static preference vector (Christakopoulou, Radlinski, and Hofmann, 2016).

Mapping must be robust to open-set behavior. When no Digital Mind explains the evidence, posterior probability is allocated to an unknown component. Uncertainty rises, the policy falls back to a simpler retrieval or exploration method, and the unexplained residual is studied offline. New Digital Minds are admitted only after they improve prospective prediction. This prevents the catalogue from expanding silently to fit every anomaly.

The mapping also distinguishes a theoretical Digital Mind from a real user record. The model stores evidence, posterior weights, uncertainty, and expiry. It should not store a claim that the person possesses a psychological trait or hidden intention. Operators and users need a plain operational description such as dynamic discovery state. The term Digital Mind names the research abstraction, not a diagnosis.

A bounded question-and-voting environment offers a practical setting for learning the mapping. Questions contain explicit answer sets, answer order can be randomized, and response latency, skipping, sequence completion, return, and voluntary satisfaction can be measured. Signals can be constructed from patterns that do not depend on topic words. The experiment can compare what a history-based ranker, a modern sequential model, and the World Engine choose for the same controlled candidate set.

$$\begin{aligned} \alpha_{i,t}(k) &\text{proportional to } \alpha_{i,t-1}(k) \int E_k(o_{i,t} | z) q_k(z_{i,t} | D_{i,1:t-1}) dz \\ q_k(z_{i,t} | D_{i,1:t}) &\text{proportional to } E_k(o_{i,t} | z_{i,t}) \int T_k(z_{i,t} | z_{i,t-1}, s_{i,t}, W_t) q_k(z_{i,t-1}) dz_{i,t-1} \\ P(y_{i,t+1} | a, D_i) &= \sum \text{over } k \alpha_{i,t}(k) \int E_k(y | z') T_k(z' | z, s(a), W_t) q_k(z | D_i) dz dz' \end{aligned}$$

Identity-to-Digital-Mind mapping is continuous and probabilistic

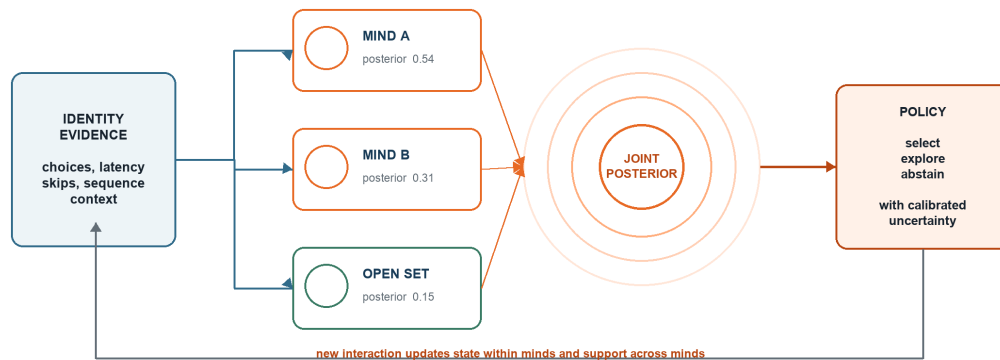


Figure 7. Identity mapping. Each observation updates both the state posterior within every supported Digital Mind and the posterior weight assigned to that model. An open-set component prevents forced classification.

8. Predictive discovery by counterfactual transition

At decision time, the system encodes each candidate as a Signal distribution and simulates possible successor states under the current posterior. The candidate receives an expected trajectory value rather than one deterministic score. Value can include immediate relevance, delayed satisfaction, diversity, information gain, user control, creator exposure, and risk penalties. The objective is explicit because deciding what to show is a normative act as well as a predictive one.

Counterfactual simulation creates a difference between recognizing a known preference and discovering possible relevance. A history-based model asks whether the candidate resembles prior positive behavior or behavior of similar users. The World Engine asks how the candidate is expected to change the present state under each plausible Digital Mind. Both sources of evidence can be combined, but they enter different parts of the computation.

The cat example can be formalized. Candidate A is another cat video. Candidate B concerns the musical Cats. Candidate C is unrelated in topic but matches a strong sequence Signal associated with exploration. A stable-preference model may favor A. A semantic model may surface B. A Digital Mind posterior indicating satiation and novelty seeking may favor C. The example illustrates competing hypotheses, not a claimed result. Randomized exposure and delayed satisfaction must determine whether the state interpretation was useful.

Temporality alone is also insufficient. After a sequence about nightlife, a history-dominant policy may continue nightlife content the following morning, while a rule-based contextual policy may substitute coffee. A transition model considers a wider candidate set whose possible relevance may arise from recovery, affirmation, humor, social connection, or novelty. The selected item need not be semantically related to nightlife or coffee. It is preferred only if its predicted successor-state distribution is superior under a declared objective and calibrated uncertainty.

The travel example supplies a stronger test. A user books a flight to Barcelona and has not searched for football. Candidate sets contain hotels, cars, restaurants, a football match, and unrelated control items. Strong funnel, multimodal sequence, latent-intent, and bandit baselines are compared with the World Engine. Success requires calibrated incremental relevance for the match among users whose current state supports it, without indiscriminate stereotyping of all travelers.

Uncertainty constrains action. A candidate with high estimated value and high epistemic uncertainty may be used for limited exploration, shown with lower frequency, or rejected when risk is high. The system should not exploit an uncertain sensitive inference simply because the possible reward is large.

$$a^* = \operatorname{argmax}_{a \text{ of } E[\sum_{h=0}^H \gamma^h U(z_{t+h}, a)] \text{ minus risk and uncertainty penalties}}$$

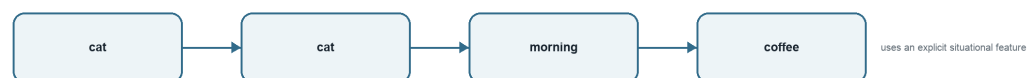
$$\text{Score}(a) = \text{expected utility} + \eta \text{ information gain} - \rho \text{ epistemic risk} - \kappa \text{ constraint violation}$$

Three discovery policies can produce different next-step trajectories

HISTORY-DOMINANT



HISTORY + CONTEXT



WORLD ENGINE HYPOTHESIS



Illustrative hypotheses only. Randomized prospective experiments must determine whether the proposed trajectory improves calibrated user outcomes.

Figure 8. Illustrative discovery trajectories. A history-dominant policy continues recorded affinity, a contextual policy responds to an explicit situational feature, and the World Engine hypothesis evaluates candidates through an estimated present state and predicted transition. The figure is a simulation design, not a performance result.

9. Relationship to world models, JEPA, and recommendation research

The nearest formal ancestor is the POMDP. A recommender observes emissions, maintains a belief state, selects an action, and receives a new observation. The central difficulty is that a useful state and transition model must be learned from policy-dependent data. Digital Minds can be interpreted as structured amortized belief models, while Signals serve as learned statistics for updating them.

World Models, PlaNet, Dreamer, and MuZero demonstrate that compact learned dynamics can support planning in latent space (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2020; Schrittwieser et al., 2020). Their environments differ from human discovery, where rewards are ambiguous, dynamics are nonstationary, and the policy changes the people and creators who generate the data. The relevant lesson is architectural: state estimation and transition prediction can be separated from immediate action scoring.

JEPA proposes prediction in representation space rather than reconstruction of every unpredictable sensory detail (LeCun, 2022; Assran et al., 2023). Signal and state encoders can be trained in the same spirit. They need not reproduce every word, pixel, or event if those details are irrelevant to future behavior. JEPA alone does not specify the Archetype hierarchy, user mapping, exposure-corrected Signal field, or discovery objective. The World Engine contributes those interfaces while leaving the encoder family open.

Model-based recommendation already learns user simulators and policies. Latent-intent models already represent hidden goals. Exploration research already studies novelty and serendipity. Ecosystem research already models coupled user and provider utilities (Chen et al., 2018; Xu et al., 2021; Boutilier, Mladenov, and Tennenholtz, 2024). The proposed contribution is incremental but coherent: shared behavioral priors, explicit Digital Mind mixtures, multivariate Signals, typed multi-timescale Functions, candidate-conditioned state transitions, and a prospective test of reduced history dependence.

The framework succeeds only if this decomposition yields measurable advantages. If a generic recurrent state model matches every benefit, the named hierarchy has not earned its complexity. If a large multimodal sequence model discovers the same cross-domain interests with better calibration, it should be preferred. The purpose of the paper is to make the comparison precise.

10. Experimental realization

The initial application is a controlled realization of the framework, not its full scope. Its unit of interaction is a visual statement or question with a bounded set of response options. Participants can vote, skip, pause, move through sequences, create moments, inspect collective responses, and enter short-lived conversations. This structure provides explicit alternatives and makes order, timing, social evidence, and collective behavior available as experimental variables.

Interface mechanics must be treated as causal variables rather than latent psychology. Answer position, option count, wording, visual prominence, displayed vote totals, response deadline, and conversation duration can alter behavior independently of content. Randomization or controlled counterbalancing is therefore required. A pattern of first-option choices is initially an observed positional regularity; only subsequent intervention can distinguish satisficing, motor habit, low deliberation, interface bias, or another explanation.

The initial system should run three policies in parallel. The first is a strong history and content baseline with multimodal and sequential features. The second is a latent-intent or state-space baseline without

named Functions or Archetypes. The third is the World Engine. All policies share candidate pools, safety filters, and comparable compute budgets. Logged propensities permit offline correction, while limited randomized traffic supports prospective inference.

The human test must answer the commercial and experiential question that simulation cannot answer. Randomized users should receive policy-balanced candidate sets, with primary outcomes defined in advance: voting probability per eligible exposure, browsing persistence, voluntary return, delayed relevance or satisfaction reports, and confidently wrong recommendations. Secondary analyses can examine value for paid users, creators, and brand participants, but these outcomes require separate measures and should not be inferred from time spent alone. Longer browsing is beneficial only when accompanied by relevance, control, and an absence of increased regret or fatigue.

Stage I uses synthetic worlds with known latent regimes. The test asks whether the model recovers predictive state, separates persistent from transient variables, and remains calibrated when the exposure policy changes. Stage II uses forward-split application logs to test Signal generalization, cold-start mapping, and natural preference shifts. Stage III randomizes answer order, semantic distance, sequence structure, and controlled exploration. Stage IV evaluates creator exposure, minority Signals, coordination, and long-term ecosystem stability.

History masking is a central test. Most of an identity's record is hidden, and the system receives only a short sequence of current interactions. A strong global model, collaborative prior, meta-learning baseline, and Archetype-Digital Mind model compete on regret, calibration, and adaptation speed. A valid reduced-history claim requires the structured model to outperform these alternatives without relying on demographic proxies.

Signal ablation tests whether behavior adds value beyond language. Candidate text and topic features are removed while response order, timing, and sequence structure remain. A complementary test randomizes semantics while preserving behavioral patterns. If performance disappears in both conditions, Signal Intelligence may be another semantic encoder. If it survives exposure randomization and predicts future behavior, the behavioral field contributes independent information.

Novel discovery tests use candidates with low semantic and collaborative similarity to history but high relevance under a controlled state variable. Metrics include calibrated acceptance, response latency, delayed satisfaction, repeat engagement, regret, and the rate of confidently wrong discoveries. Immediate clicks alone are insufficient. A policy that increases response while reducing satisfaction or autonomy has failed.

Causal claims require interventions. Retrospective associations can identify useful prediction but cannot establish that a Signal or candidate caused a transition. Randomized and quasi-experimental variation is used where safe. The model logs exposure propensity, posterior summaries, uncertainty, predicted outcomes, and delayed feedback so that policy-induced patterns can be audited.

10.1 Falsifiable claims and failure criteria

Claim	Test	Primary evidence	Failure criterion
Reduced history dependence	Cold start and history masking	Calibration, regret, adaptation speed	No gain over strong priors or sequence models
Signal Intelligence adds structure	Prospective pattern tests	Predictive lift after exposure correction	Signal vanishes under randomization

Claim	Test	Primary evidence	Failure criterion
Digital Minds improve shift	Goal, context, and preference change	Detection delay and log loss	Mixture is slower or less calibrated
Rollouts improve discovery	Low-similarity candidate sets	Delayed satisfaction and regret	More clicks but worse later outcomes
World State captures externality	Creator and user feedback loops	Exposure quality and concentration	Simulation fails policy transfer

Table 2. Claims are supported only through prospective evidence against capacity-matched baselines.

11. Governance, autonomy, and limits

A system that predicts latent behavioral dynamics can increase relevance and manipulation at the same time. Governance is therefore part of the computational specification. Objectives must be approved explicitly rather than inferred from engagement. Safety, privacy, and rights constraints cannot be traded away through one scalar reward.

Sensitive attributes should not be inferred merely because they improve prediction. Health, political belief, religion, sexual orientation, and other protected or intimate characteristics require prohibited-inference policies. Behavioral clusters are tested for proxy leakage. Data are minimized, retained for limited periods, access controlled, and associated with user correction and deletion mechanisms.

Uncertainty must be operational. The policy abstains when a candidate rollout is out of distribution or when Digital Mind support is diffuse. A simple and auditable recommender is preferable when the structured model is poorly calibrated. High-impact allocation requires human review and separate governance.

The system should preserve user agency. It can expose why a recommendation appeared in ordinary terms, provide controls for repetition and exploration, permit reset or correction, and avoid covertly optimizing emotional vulnerability. The goal is not to make a platform irresistible. It is to discover information that a person is more likely to find relevant or valuable under transparent objectives.

The neuroscience analogy has a strict boundary. Platform interaction cannot validate a biological theory of mind. A Digital Mind remains a computational record associated with evidence and uncertainty. It must not be presented to users, operators, or investors as consciousness, diagnosis, or a faithful replica of a person.

12. Discussion

The World Engine proposes a new generation of discovery architecture rather than one new neural layer. It begins from a representational claim: digital actions are sparse emissions from processes that develop over time. It then introduces a hierarchy for sharing transition structure, a Signal field for detecting meaningful event patterns, a probabilistic bridge from Digital Minds to real identities, and a counterfactual mechanism for selecting information.

The most important novelty claim is reduced dependence on identity-specific history. That claim is easy to overstate. The model learns from historical population data, and it uses current interaction evidence. Its proposed advantage is the transfer of functional structure across identities and the

separation of present state from accumulated profile. Cold-start and history-masking tests decide whether that advantage exists.

The framework also changes the meaning of discovery. Similarity remains useful, but it is not sufficient. A candidate may matter because it is likely to move an uncertain state toward interest, understanding, novelty, task completion, or another declared outcome. This makes recommendation a problem of transition under uncertainty rather than only a problem of ranking affinity.

Several outcomes would narrow the thesis. Archetypes may provide no advantage over universal meta-learning. Digital Mind components may be nonidentifiable. Signal detectors may overfit exposure artifacts. Candidate rollouts may compound model error. The full architecture should not survive by definition. Each component remains only if prospective evidence justifies it.

If the proposed structure succeeds, the implication extends across social feeds, streaming media, commerce, programmatic advertising, education, and digital assistance. Each domain selects information for people whose present state is only partially observed. A discovery system that models the transition from current state to possible relevance could complement language and history with a third source of intelligence: a calibrated model of behavioral dynamics.

The immediate task is empirical. Build controlled Digital Minds, map them to real identities without demographic shortcuts, detect Signals that survive randomized exposure, and compare the resulting policy with the strongest practical baselines. The World Engine becomes a scientific contribution only through that sequence of tests.

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Appendix A. Formal specification

A World Engine is a tuple $W = (Z, O, A, S, G, T, E, R, U)$, containing latent states, observations, actions, Signals, a relation graph, transition and emission models, a routing operator, and objective functions. At time t the inference model estimates $q(Z_t | O \text{ up to } t, A \text{ before } t, C \text{ up to } t)$. The action-conditioned transition predicts $p(Z_{t+1}, W_{t+1} | Z_t, A_t, S_t, G_t)$. The emission model closes the generative loop.

A practical training objective combines future-event prediction with transition consistency, task loss, calibration, and regularization. Contrastive or joint-embedding losses can replace full observation reconstruction when nuisance detail dominates. Identifiability is not assumed. States are treated as equivalent when they induce the same distribution over future tests relevant to the decision problem.

Multi-timescale dynamics separate slowly varying constraints, intermediate memory and goals, and rapidly changing context and attention. Continuous-time or selective state-space implementations can learn decay constants. Event-driven implementations update only affected components. Missing observations require explicit masks and elapsed-time features.

Relational propagation uses bounded graph messages with provenance. Messages represent observable influence, not access to another person's private state. Temporal graph splits prevent leakage from future interactions.

$$\text{Loss} = L_{\text{future}} + \beta L_{\text{transition}} + \lambda L_{\text{task}} + \tau L_{\text{calibration}} + \omega L_{\text{complexity}}$$

Appendix B. Reference architecture

One implementation combines a multimodal event encoder, a library of temporal and graph Signal detectors, a hierarchical mixture prior, a selective state-space filter for per-identity updates, retrieval over episodic summaries, an ensemble transition model for epistemic uncertainty, and a constrained model-predictive controller. Training proceeds through self-supervised event prediction, Signal validation, supervised outcome calibration, intervention-aware transition learning, and conservative policy optimization.

For each event, the system records content and question representation, answer order, exposure mechanism, timestamp, response, context, and propensity. It retrieves relevant memory, updates the posterior within each Digital Mind, updates mixture support across Digital Minds, simulates candidate successor states, estimates outcome distributions, and applies constraints. It abstains or falls back when uncertainty is high.

The implementation maintains strict separation between research nomenclature and operational records. Digital Mind identifiers are model component IDs. Archetypes are mixture priors. Identity records contain observations and posterior weights. No component is described as a biological brain or psychological diagnosis.

Appendix C. Minimum evaluation protocol

Element	Minimum requirement
Data	Temporally ordered exposures, interactions, context, delayed outcomes, and randomized exploration where lawful.
Baselines	Collaborative filtering, GRU4Rec, SASRec, transformer and state-space recommenders, latent-intent models, contextual bandits, and model-based RL.
Splits	Forward time, user cold start, item cold start, policy shift, context shift, preference shift, and graph shift.
Metrics	Log loss, Brier score, calibration error, NDCG, regret, adaptation delay, delayed satisfaction, diversity, exposure concentration, privacy leakage, and compute.
Ablations	Remove Archetypes, typed Functions, memory, relational propagation, uncertainty, Signal families, and rollouts independently.
Reporting	Confidence intervals, effect sizes, seed variance, compute budgets, calibration plots, negative results, and cases where simpler models win.

Table 3. Minimum protocol for a credible empirical claim.

Appendix D. Editorial and terminology rules

- World Engine names the complete computational framework.
- Signal Intelligence names detection and validation of predictive patterns in the event field.
- An Archetype is a shared prior over platform-relevant transition dynamics.
- A Digital Mind is one computational transition hypothesis.
- An identity is a real or synthetic actor associated with observations and posterior weights.
- A Function is a reusable state-transition operator.

The paper avoids claims that the system reads a brain, discovers a true private intention, proves biological determinism, or reproduces a person. Biological evidence motivates abstractions. Behavioral and platform experiments validate or reject the computational model.